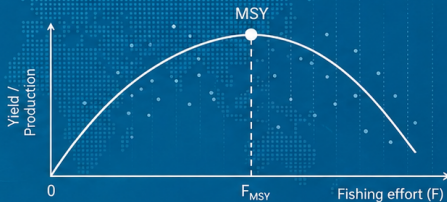




Food and Agriculture
Organization of the
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FLicc: Length interval catch curve analysis for multi-gear applications in FLR

Concept, motivation, indicators and advice



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Motivation

- ▶ FLicc is an **user-friendly TMB implementation in the FLR** and an extension of both the multi-gear length-interval catch-curve framework developed in fishblicc and length-based spawning potential ratio estimation in LBSRP.
- ▶ It retains the biological core of the original equilibrium model, but adds:
 - ▶ **fast penalized-likelihood estimation in TMB**
 - ▶ **multi-year fitting**
 - ▶ **FLR integration**
 - ▶ **equilibrium and indicator tools**
- ▶ The original motivation is simple: many real fisheries are **multi-gear**, but many length-based methods, such as LBSRP assume a **single gear with logistic selectivity**.

What is new relative to fishblicc and LBSRP?

Practical extensions in FLicc

- ▶ **Multi-year estimation framework for multi-gear applications**
- ▶ **Random-walk structure in annual fishing mortality**
- ▶ **Alternative population and observation models**
- ▶ **FLR object outputs**
- ▶ **Indicator development**, including LBI_{spr}
- ▶ **Fast TMB implementation** designed for MSE applications

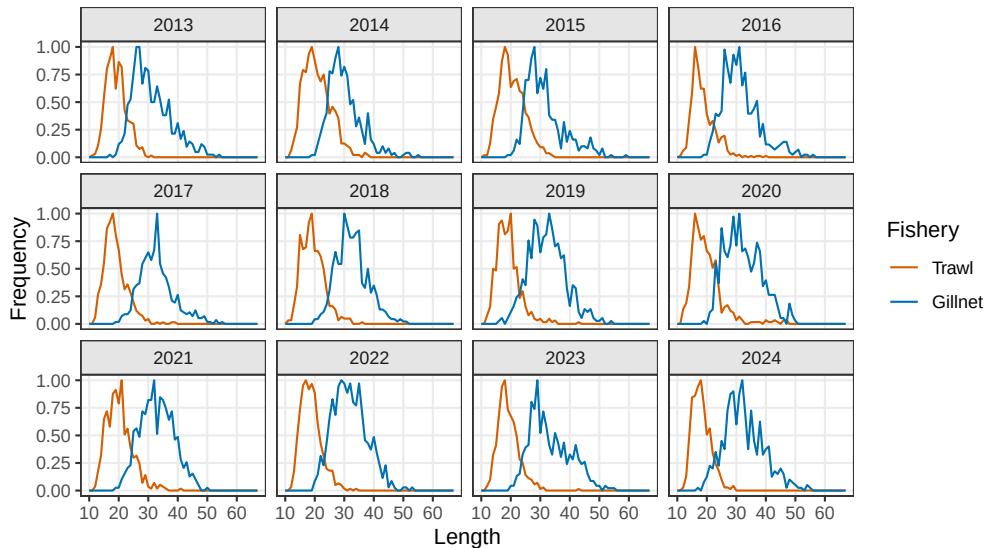
These features make FLicc attractive both for **method development** and for **applied workflows**.

Why multi-gear matters

Long-lived species and mixed fleets

- ▶ In longer-lived species, mixed fleets often exploit **different parts of the size distribution**.
- ▶ This creates a core identifiability problem:
 - ▶ the declining right tail of the catch-at-length distribution may reflect
 1. real mortality, and/or
 2. the descending limb of gear selectivity.
- ▶ Joint fitting across gears helps separate:
 - ▶ **fishing mortality**, and
 - ▶ **selectivity shape**.

Simulated splendid alfonsino length samples by gear



Relationship to LBSPR

A reproducible and extensible bridge

- ▶ FLicc is closely related to **LBSPR**.
- ▶ Under the special case of:
 - ▶ **single fleet**
 - ▶ **logistic selectivity**
 - ▶ **multinomial likelihood**
- ▶ FLicc reproduces **LBSPR-type inference**, while extending the framework to more realistic fleet structures.

$$\frac{Z}{K} = \frac{M + F}{K}$$

- ▶ This length-structured per-recruit logic underpins both frameworks.

Link to fishblicc

From Stan to FLR/TMB

- ▶ fishblicc introduced a **Bayesian multi-gear catch-curve framework** for length data.
- ▶ FLicc keeps the core equilibrium interpretation, but adds:
 - ▶ **speed**
 - ▶ **joint multi-year fitting**
 - ▶ **FLR compatibility**
 - ▶ **equilibrium reference curves**
 - ▶ **indicator workflows**
- ▶ GitHub: github.com/Henning-Winker/FLicc

Core model idea

Catch-at-length under growth and mortality

$$c_i = \frac{F_i}{Z_i}(S_i - S_{i+1})$$

where:

- ▶ S_i = survival to the lower bound of length interval i
- ▶ F_i = fishing mortality at length
- ▶ $Z_i = M_i + F_i$ = total mortality at length

For gear j , catch per recruit is:

$$C_j = \sum_{i=1}^N \frac{F_{ij}}{Z_i}(S_i - S_{i+1})$$

Multi-gear decomposition

Shared population, gear-specific selectivity

$$Z_i = M_i + \sum_j F_j \text{ sel}_{ij}$$

- ▶ All gears observe the **same underlying population**.
- ▶ Each gear samples that population through a **different selectivity function**.
- ▶ The joint likelihood constrains mortality and selectivity simultaneously.

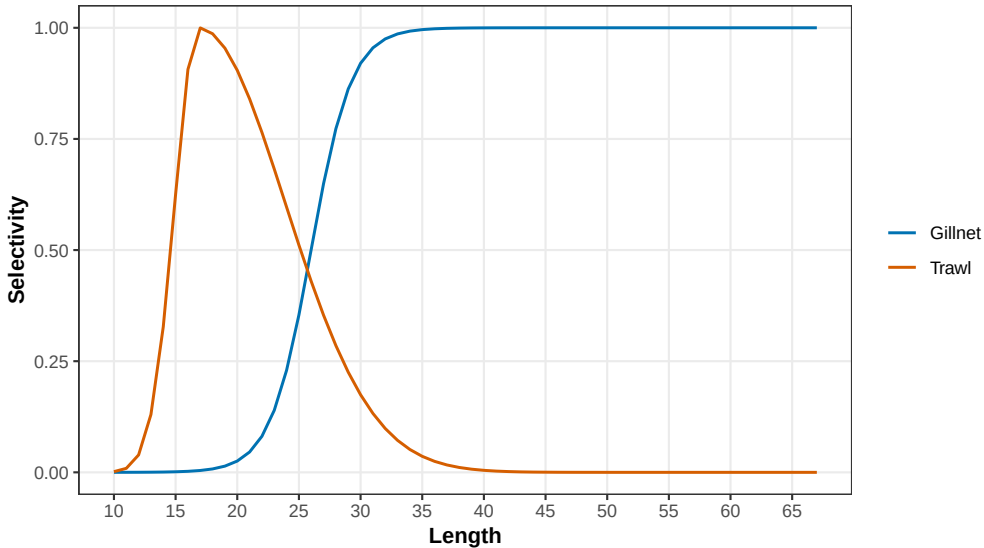
Selectivity options

Why flexibility matters

FLicc currently supports:

- ▶ **logistic**
- ▶ **normal**
- ▶ **double-sided normal**

This is especially important in fisheries where some gears are **dome-shaped**.



- A multi-gear framework is particularly useful when at least one gear still samples the larger fish.

Population and observation models

Population model + observation model separation

Default configuration:

```
settings = list(pop_model = "gtg", obs_model = "mn")
```

- ▶ `pop_model = "gtg"`: growth-type-group approximation
- ▶ `obs_model = "mn"`: multinomial observation model

This default is typically **much faster** than the original fishblicc-style gamma integration workflow while retaining the key length-based structure.

Population model options

GTG versus gamma

▶ GTG (default)

- ▶ efficient recursion over growth-type groups
- ▶ recommended for most applications

▶ gamma

- ▶ closer to the original fishblicc integration approach
- ▶ slower, but useful for replication / comparison

$$N_{i+1} = N_i \exp(-Z_i \Delta t_i)$$

- ▶ The model is still rooted in **equilibrium length structure under growth and mortality**.

Observation model options

Flexible composition likelihoods

- ▶ **Multinomial**

$$\mathbf{y} \sim \text{Multinomial}(n, \pi)$$

- ▶ **Dirichlet-multinomial** for overdispersion

- ▶ **Negative binomial** for count-based flexibility

- ▶ In practice:

- ▶ `mn` is fast and stable

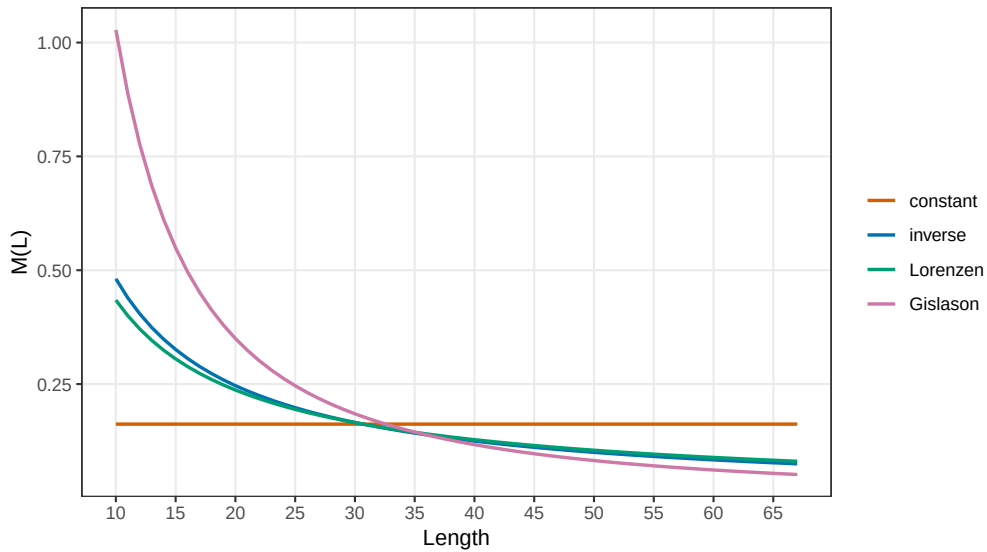
- ▶ `dm` is attractive when there is composition overdispersion

- ▶ `nb` can be useful, but may be less stable

Life-history input

- ▶ FLicc uses an FLStockLen-style structure to hold:
 - ▶ growth
 - ▶ maturity
 - ▶ mortality-at-length
 - ▶ Constant
 - ▶ Inverse length scaling
 - ▶ Lorenzen
 - ▶ Gislason
 - ▶ length-weight information

This keeps the workflow fully embedded in **FLR**.

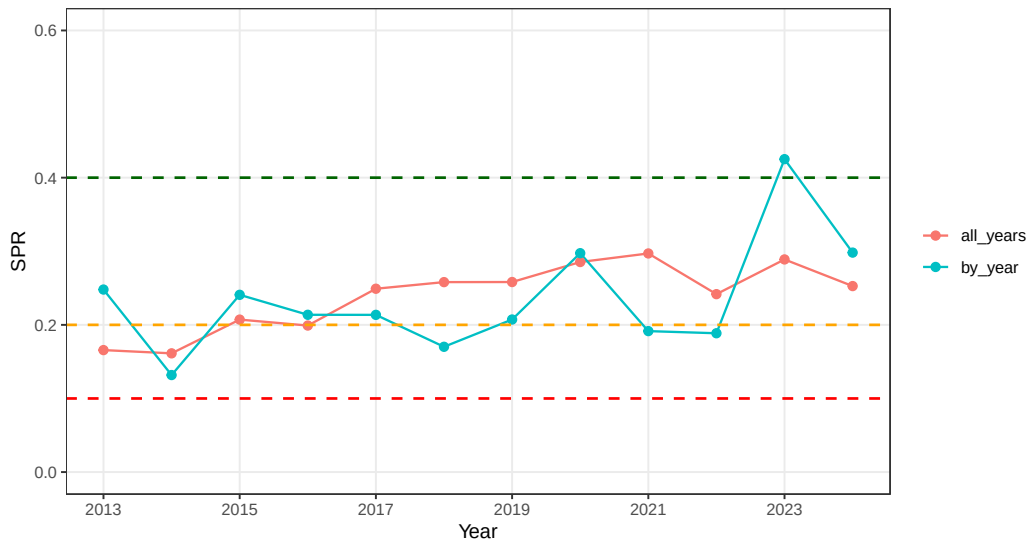


Fitting across years

Borrowing strength across annual data

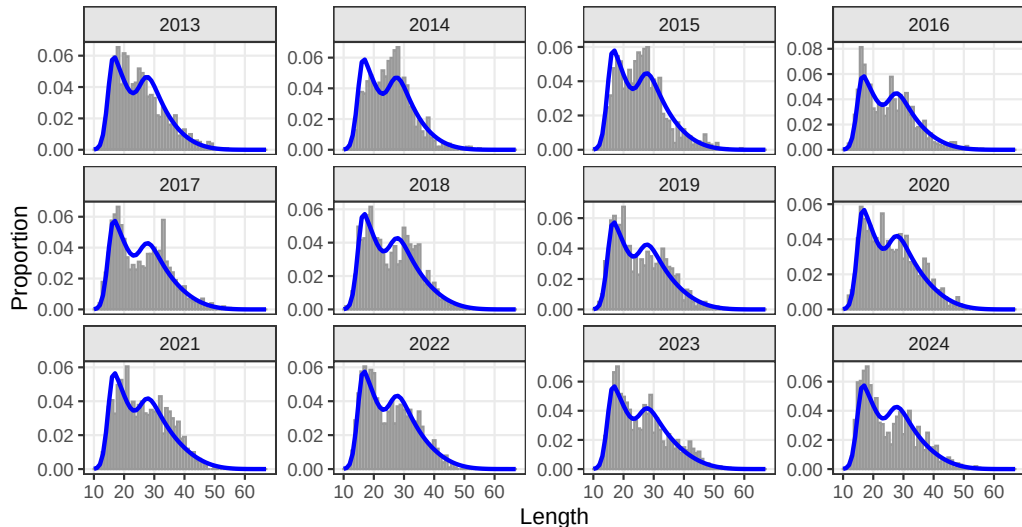
- ▶ FLicc can fit:
 - ▶ **all years jointly**, or
 - ▶ **each year separately**
- ▶ Multi-year fitting improves identifiability of:
 - ▶ life-history and selectivity parameters
 - ▶ year-specific fishing mortality

Spawning potential ratio

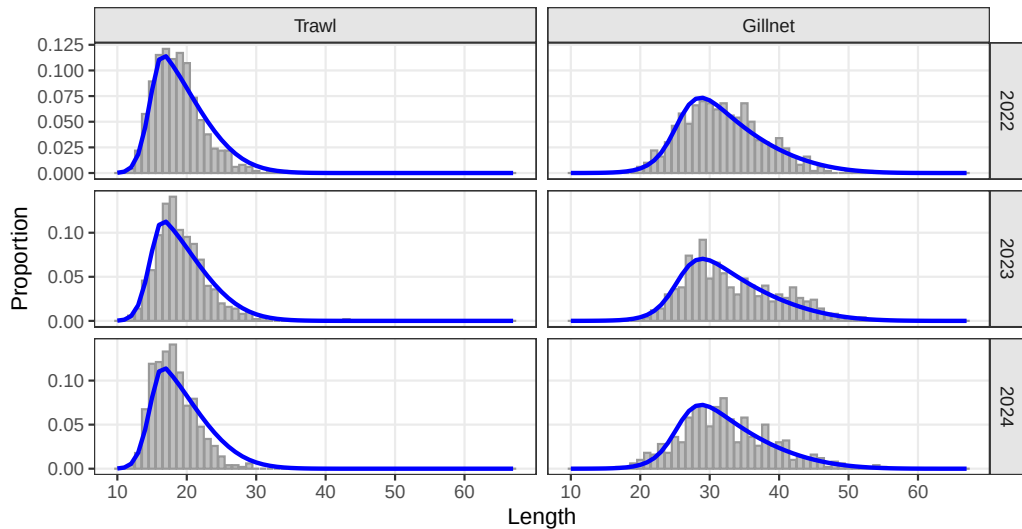


Observed versus fitted length compositions

Observed and fitted length compositions

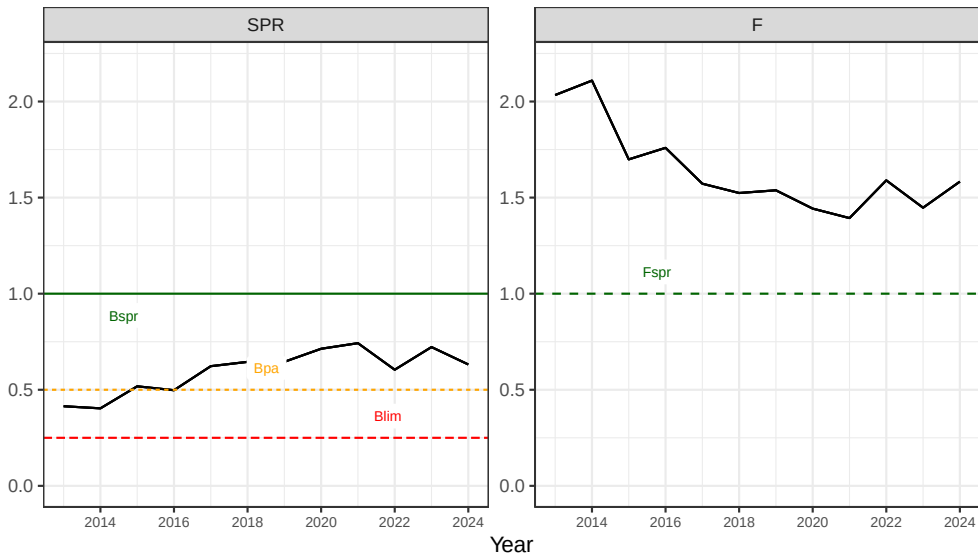


Observed and fitted length compositions



- ▶ Key diagnostic questions:
 - ▶ Are the dominant modes reproduced?

Convert to FLStock for length-based status advice

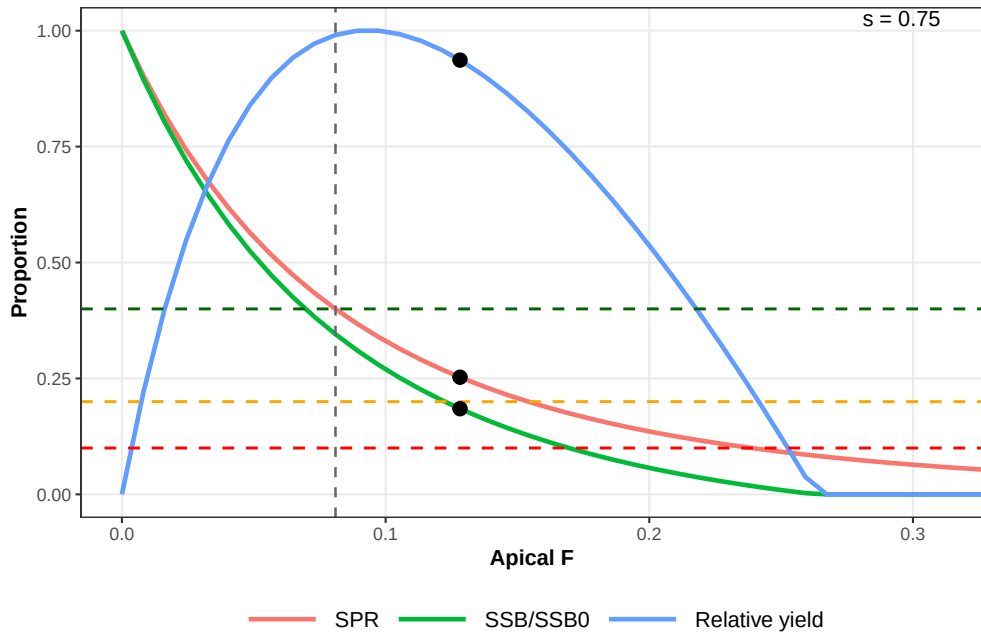


Equilibrium reference points

- ▶ `eqstklen()` provides equilibrium relationships among:
 - ▶ fishing mortality
 - ▶ SPR
 - ▶ relative biomass
 - ▶ relative yield

This is useful for **communication**, **reference point exploration**, and **stock status classification**.

Equilibrium curves



LBIspr motivation

Why a structure-based indicator?

- ▶ CPUE can be extremely valuable, but in many fisheries it may be:
 - ▶ unavailable
 - ▶ biased by targeting effects
 - ▶ inconsistent through time
 - ▶ difficult to standardize across fleets
- ▶ Length data are often more widely available.

LBIspr was developed to provide a **structure-based index** that can:

- ▶ substitute for a relative abundance proxy when CPUE is weak or absent
- ▶ corroborate CPUE trends where CPUE exists

LBI_{spr} concept

Observed structure versus expected target structure

$$LBI_{SPRx,t} = \frac{P_t(L \geq L_{SPRx})}{P_{SPRx}}$$

where:

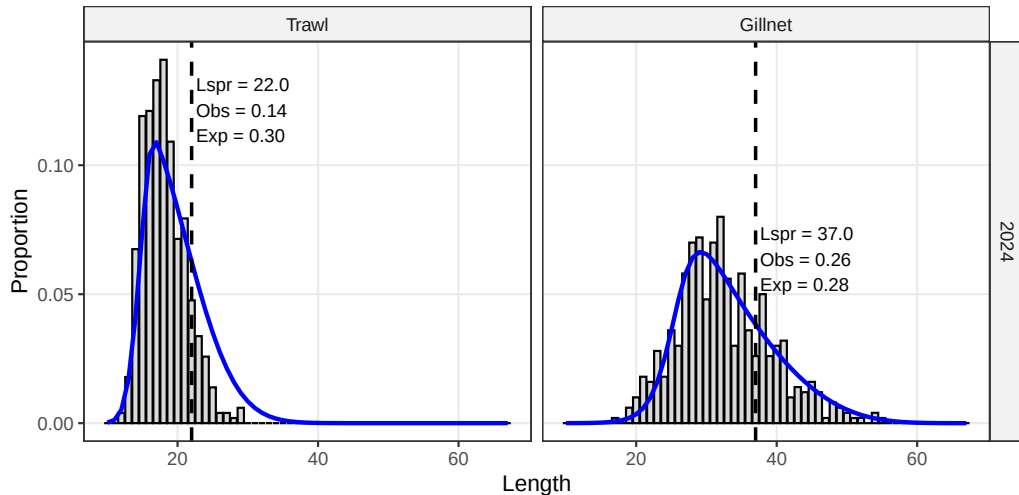
- ▶ L_{SPRx} is a large-fish threshold linked to a target SPR level
- ▶ $P_t(L \geq L_{SPRx})$ is the observed proportion above that threshold
- ▶ P_{SPRx} is the expected equilibrium proportion above that threshold under the target SPR

Interpretation:

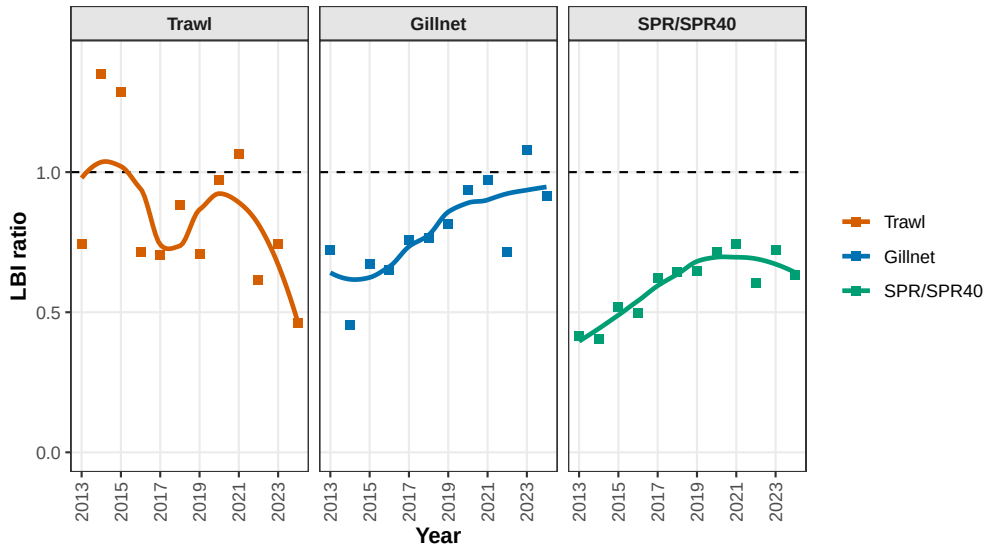
- ▶ < 1 : fewer large fish than expected under the target structure
- ▶ ~ 1 : close to target
- ▶ > 1 : more large fish than expected

Observed vs expected length composition at Fspr40

Expected curve from `nf_flicc()` at SPR40; dashed line shows Lspr threshold (thresh = 0.75)



LBI spr indices



LBI_{spr} and CPUE

Complement or substitute

Potential roles of LBI_{spr}:

1. **Corroborate CPUE**

- ▶ if both indicators move in the same direction, confidence improves

2. **Proxy for relative abundance / exploitation signal**

- ▶ especially in long-lived species where large fish are informative

3. **Support interpretation of selectivity and structure changes**

- ▶ LBI_{spr} should not be treated as identical to CPUE.

- ▶ It is better viewed as a **structure-based index with CPUE-like utility.**

MSE applications

From assessment output to management testing

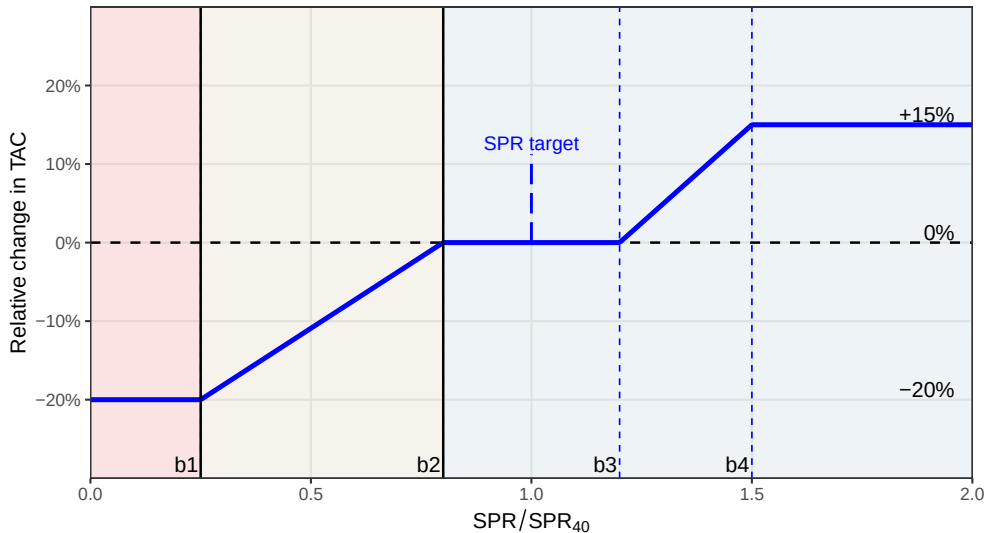
Length-based indicators can be embedded directly in **Management Strategy Evaluation (MSE)**.

Examples:

- ▶ LBSPR / FLicc outputs can support **rule-based harvest control rules**
- ▶ candidate control variables may include:
 - ▶ **SPR** or relative SPR-to-target
 - ▶ **LBI_{spr}**
 - ▶ **mean length** L_{mean}
 - ▶ empirical ratios such as L_{mean}/L_M

This is particularly attractive for **data-limited fisheries**.

Example of HCR relative SPR-to-target



ICES Category 2 and 3 context

Link to empirical advice rules

- ▶ Under **ICES Category 2 and 3** guidelines, empirical and indicator-based approaches can be used where full analytical assessments are not available.
- ▶ This creates a natural entry point for:
 - ▶ **length-based indicators**, and
 - ▶ **SPR-based control variables**

A useful applied route is the **rule-based framework discussed by Simon Fischer**, where fishing pressure or catch advice may be linked to length-based indicators.

Example MSE note

Rule-based harvest control rules

Candidate control variables in MSE could include:

- ▶ **SPR-based F proxies** from LBSPR / FLicc
- ▶ empirical indicators such as:
 - ▶ L_{mean}
 - ▶ L_{mean}/L_M
- ▶ LBIspr as a structure-based abundance / exploitation proxy

Conceptually, an MSE could test whether LBIspr:

- ▶ tracks stock status sufficiently well,
- ▶ stabilizes advice when CPUE is poor,
- ▶ and improves robustness of empirical harvest rules.

GitHub and reproducibility

Reproducible workflow

- ▶ Repository: **github.com/Henning-Winker/FLicc**
- ▶ Includes:
 - ▶ worked examples
 - ▶ test scripts
 - ▶ equilibrium and plotting functions
 - ▶ FLR integration

A central aim is to provide a **reproducible extension of LBSPR and fishblicc concepts** within an FLR/TMB workflow.

Take-home messages

Summary

- ▶ FLicc extends **fishblicc** into a fast **FLR/TMB** workflow.
- ▶ It provides a reproducible bridge to **LBSPR** under the single-gear logistic special case.
- ▶ Its main strength is realistic inference for **multi-gear fisheries**.
- ▶ LBI spr provides a promising **structure-based indicator** that can:
 - ▶ complement CPUE,
 - ▶ substitute for CPUE when needed,
 - ▶ and be tested in **MSE / empirical HCR frameworks**.

References

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